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Easily accessible linear and hyperbranched polyesters as solid polymer electrolytes

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ABSTRACT

Herein we report synthesis, characterization, thermal properties, ionic conductivity and oxidative stability of novel linear and hyperbranched polyesters. Tri-acrylate ester of commercially available 1,1,1-Tris(hydroxymethyl)propane (**M1**) was used as the primary building block for the synthesis of linear and hyperbranched polymers. A3 + B2 type polycondensation between **M1** and dithiols of two different chain lengths (C3, C6) by 100 % atom efficient thiol-acrylate Michael addition reaction, followed by consumption of the unreacted acrylate esters (in the linear or terminal units) with 1-butane thiol produced two hyperbranched polymer namely **HB-P1** and **HB-P2**. For synthesis of linear analogues, one of the equivalent acrylate esters was reacted first with 1-butane thiol and the resulting di-acrylate monomer was polymerized with the C3 and C6-dithiols, producing linear polymers **L-P1** and **L-P2**. All the polymers showed molecular weight (M_n) in the range of 5000–6000 gmol^{-1} with low dispersity. TGA analysis revealed sufficient thermal stability of the polymers for the application as solid polymer electrolyte. All the polymers are amorphous, showing only a glass transition in the range of ~ -45 °C (C3 spacer) to −50 °C (C6 spacer) and no crystallization peak. In presence of lithium bis(trifluoromethanesulfonyl) imide (LiTFSI) salt the T_g increased by ~ 10 °C. Solid polymer electrolytes were prepared by incorporating LiTFSI at a molar ratio of polymer carbonyl units to lithium of C=O/Li of 10 and ionic conductivity (σ) was evaluated by electrochemical impedance spectroscopy (EIS). Moderate values in the range of 10^{-7} S cm^{-1} were obtained at 25 °C with slightly higher values for the linear polymers. The ionic conductivity increased up to ~ 2 orders of magnitude at higher temperatures. The oxidative stability of the polymer electrolytes against lithium-metal electrodes revealed onset potential for the first degradation in the range of 3.8 V, indicating moderate stability that might be suitable for testing applications in all-solid-state lithium-metal batteries using sulfur or lithium iron phosphate cathodes.

1. Introduction

Lithium-ion batteries (LIB) are of high demand to meet sharply increasing need for energy storage technologies suitable for electric vehicles or other electronics [1,2]. Conventionally such batteries contain flammable liquid electrolytes which is not desirable due to the growth of Li dendrite and serious safety related concerns if used in combination with metallic lithium anodes [3]. Hence there is a tremendous research interest in solid polymer electrolytes (SPE) for all solid-state battery application [4,5] and till date poly(ethylene oxide) (PEO) is considered to be the benchmark candidate as solid polymer electrolyte for LIB [6,7]. High degree of crystallinity, low ionic conductivity at room temperature, low lithium transport number, and not so high electrochemical stability are major drawbacks of PEO [8]. To address these issues, in the recent past other synthetic polymers "beyond PEO" have been explored as solid electrolyte for LIB [9–12]. Amongst them, polyesters are particularly promising considering weak Li coordination resulting in high lithium transport number compared to PEO. From a few recent reports, [13–15] including recent work by our group, [16–18] it is evident that some of the pressing problems in the solid polymer electrolytes may be addressed by appropriate structural fine tuning in easily accessible polyester scaffolds, which also offer the

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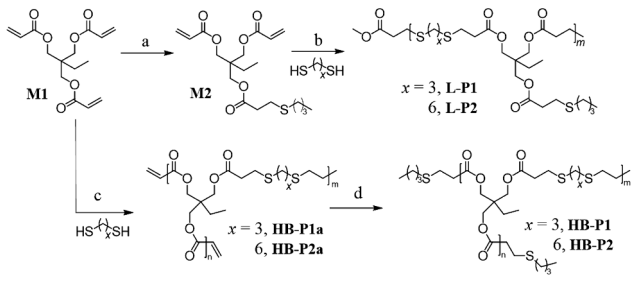
additional advantage of biodegradability. Inspired by these recent promising results, we wanted to explore new polyesters for LIB. Polyesters are generally synthesized by transesterification reaction which requires high temperature and continuous removal of the volatile alcohol to push the reaction in the forward direction. This may not be the ideal scenario for exploring functional polymers in which structure–property relationship studies often requires comparing the properties of a series of polymers. Hence mild and versatile synthetic methods [19] may enhance the scope of polyesters in solid polymer electrolyte of other applications. We had established a methodology for synthesis of polyesters by thiol-acrylate Michael addition reaction between diacrylate and dithiol monomers [20].

Although it produces polyesters with a sulphur atom present at the β-position to the ester group in every repeating unit, the polymerization reaction undergoes at room temperature and the reaction is 100 % atom efficient as there is no byproduct during the reaction. In this manuscript we report synthesis of linear and hyperbranched polyesters (**L-P1**, **L-P2**, **HB-P1**, **HB-P2**, Scheme 1) by thiol-acrylate Michael addition reaction and study the effect of structural nuances on thermal properties, lithium-ion conductivity and electrochemical stability.

2. Experimental

Materials and methods: Solvents and reagents were purchased from commercials sources and purified by prescribed protocols. Trimethylolpropane triacrylate (**M1**), 1-butanethiol, 1,3-propanedithiol, 1,6-hexanedithiol, methyl-acrylate, Me_2PPh, acetonitrile (99.8 %, anhydrous) and Lithium bis(trifluoromethane)sulfonimide (LiTFSI, 99.9% ^{19}F NMR) were purchased from Sigma Aldrich. The Li-salt was dried at 60 °C under vacuum overnight before storing in an Ar-filled glovebox. Chloroform (99 %) was retrieved from VWR chemicals. CR2032-type coin cell parts (cases, stainless steel (StSt) spacers (500 μm thickness), stainless steel wave springs) were bought from PI-KEM limited. Teflon® sheets (500 μm thickness) and Kapton® sheets (135 μm thickness) were purchased from Goodfellow Cambridge Ltd. Lithium metal ribbon (purity 99.9 %, thickness 380 μm, width 23 mm) was retrieved from Sigma Aldrich, lithium electrodes were punched from the ribbon using a 14 mm diameter punching die inside the glovebox. Spectroscopic grade solvents were used for all physical studies. ^1H NMR spectra were recorded in a Bruker DPX-400 MHz spectrometer the peak positions were calibrated using TMS as an internal standard. Molecular weight of the polymer was estimated in THF (1.0 mg/mL) solvent at 30 °C with respect to poly (methyl methacrylate) (PMMA) standards in a Waters GPC machine equipped with a 515 HPLC pump, Waters 2414 RI detector, and HSPgel HT 4.0/HSPgel HT 2.5 columns connected in a series.

Synthesis of L-P1: Monomer **M2** (1 g, 2.6 mmol), propane-1,3-dithiol (308 mg, 2.8 mmol) and methyl acrylate (47 mg, 0.546 mmol) were dissolved in anhydrous and degassed dimethylacetamide (DMAc) and placed in an ice-bath. To this reaction mixture Me_2PPh (0.03 mmol) was added and stirred at room temperature for 1 h under inert atmosphere. The polymer was collected by precipitation from cold excess diethyl ether and isolated as a colourless viscous liquid. Yield = 800 mg (~61 %). ^1H NMR ($CDCl_3$, 300 MHz, TMS): δ (ppm) = 4.05 (d, 6H), 3.69 (s, 6H), 2.75 (m, 6H), 2.64 (m, 12H), 1.85 (m, 1H), 1.50 (m, 6H), 0.91 (m, 6H). End group analysis from ^1H NMR revealed molecular weight of **L-P1** to be 6500 g/mol. Molecular weight obtained from SEC in THF is 5500 g/mol (Đ = 1.2).



Scheme 1. Synthesis of the linear and hyperbranched polymers studied in this paper. Reagents and conditions: (a) 1-butanethiol, Me_2PPh, THF, 0 °C-rt, 1 h; (b) Methyl-acrylate, Me_2PPh, DMAc, 0 °C-rt, 1 h; (c) Me_2PPh, DMAc, 0 °C-rt, 2 h; (d) 1-butanethiol, Me_2PPh, THF, 0 °C-rt, 2 h.

Synthesis of L-P2: For the preparation of **L-P2**, the similar procedure as for the **L-P1** synthesis was followed. In this case hexane dithiol was utilized for the thiol acrylate polymerization reaction. Yield = 900 mg (~64 %). ^1H NMR ($CDCl_3$, 300 MHz, TMS): δ (ppm) = 4.05 (d, 6H), 2.75 (t, 6H), 2.60 (t, 6H), 2.49 (t, 6H), 2.34 (t, 4H), 1.85 (m, 1H), 1.58 (b, 6H), 1.39 (b, 4H), 0.91 (m, 6H). End group analysis from ^1H NMR revealed molecular weight of **L-P2** to be 7500 g/mol. Molecular weight obtained from SEC in THF is 5000 g/mol (Đ = 1.2).

Synthesis of HB-P1a: Commercially available trimethylolpropane triacrylate (**M1**) (0.5 g, 1.68 mmol) and monomer propane dithiol (0.134 g, 1.2 mmol) were dissolved in 3 mL of degassed dimethylacetamide (DMAc) in inert atmosphere and was placed in an ice-bath. Then catalytic amount of Me_2PPh (0.03 mmol) was added under inert atmosphere to this reaction mixture and it was stirred for 2 h at rt. The reaction was subsequently stopped and the polymer (**HB-P1a**) was collected by precipitation from excess cold diethyl ether and isolated as highly viscous colourless liquid after drying under vacuum. Yield: 400 mg (~63 %). ^1H NMR ($CDCl_3$, 400 MHz) δ (ppm): 6.44 (d, 1H), 6.14 (m, 1H), 5.88 (d, 1H), 4.16–4.07 (m, 18H), 2.78 (b, 2H), 2.64 (b, 4H), 1.87 (b, 2H), 1.52 (b, 2H), 0.91 (b, 3H).

Synthesis of HB-P2a: For the preparation of **HB-P2a**, similar reaction procedure was followed as in the case of **HB-P1a**. Here, the hexane dithiol was used as the dithiol monomer for thiol acrylate polymerization reaction. Yield = 500 mg (~70 %). ^1H NMR ($CDCl_3$, 400 MHz) δ (ppm): 6.38 (d, 1H), 6.12 (m, 1H), 5.87 (d, 1H), 4.15–4.05 (m, 18H), 2.75 (b, 2H), 2.60 (b, 4H), 2.51 (b, 2H), 1.6 (b, 8H), 1.38 (b, 2H), 0.90 (b, 3H).

Synthesis of HB-P1: **HB-P1a** (0.3 g) and 1-butane thiol (0.435 g, 4.83 mmol) was dissolved in anhydrous THF under an inert atmosphere and placed in an ice bath. Subsequently, catalytic amount of Me_2PPh (0.03 mmol) was added to this reaction mixture maintaining the inert atmosphere. The reaction mixture was stirred for 2 h at rt. The reaction was stopped then and the polymer (**HB-P1**) was isolated by precipitation from excess cold diethyl ether. The purified polymer was vacuum dried and collected as a colourless viscous liquid. Yield = 310 mg (~90 %). ^1H NMR ($CDCl_3$, 400 MHz) δ (ppm): 4.05 (s, 18H), 2.76 (b, 2H), 2.62 (b, 4H), 2.52 (b, 2H), 1.85 (b, 2H), 1.48 (b, 2H), 0.91 (b, 3H). Molecular weight (Mw) from SEC: 5500 g/mol (Đ =1.18).

Synthesis of HB-P2: For the preparation of **HB-P2**, similar reaction procedure was followed as in the case of **HB-P1**. Yield = 420 mg (~95 %) ^1H NMR ($CDCl_3$, 400 MHz) δ (ppm): 4.08 (m, 18H), 2.78 (b, 2H), 2.63 (b, 2H), 2.54 (b, 2H), 1.61 (b, 8H), 1.41 (b, 2H), 0.94 (b, 3H). Molecular weight (Mw) from SEC: 6300 g/mol (Đ =1.25).

Thermogravimetric Analysis (TGA): TGA thermograms were recorded on a Mettler Toledo TGA/DSC 3. Samples (ca. 5–10 mg) were placed in aluminum oxide ceramic crucibles (70 μL) and heated from 30 to 700 °C at a heating rate of 10 K min^{-1} under nitrogen atmosphere.

Differential Scanning Calorimetry (DSC): DSC thermograms were obtained using a Mettler Toledo DSC 3. The samples were sealed in aluminum crucibles. The temperature program comprised two cycles of cooling down to − 70 °C and subsequent heating up to 150 °C at a cooling/heating rate of 10 K min^{-1}. For the third cycle, a cooling/heating rate of 40 K min^{-1} was chosen. Glass transition temperatures (T_g) are derived as inflection points of the transition temperature in the second heating curve.

Electrochemical Measurements: Electrochemical measurements were conducted using a VMP300 potentiostat (BioLogic) equipped with EIS boards. Before any measurement coin cells were tempered at 70 °C at open-circuit voltage (OCV) for at least 12 h. The temperature was

controlled using a climate chamber Vötsch VT4002.

Electrolyte Preparation: The electrolytes were prepared by dissolving the polymer in chloroform (ca. 0.75 mL per 100 mg of polymer). Lithium-bis(trifluoromethylsulfonyl)imide (LiTFSI) was weighed inside a argon-filled glovebox. The molar concentration of lithium salt was 10:1 with respect to the carbonyl groups of the polymer repeating unit to the number of lithium cations (C=O/Li 10). The LiTFSI is dissolved in acetonitrile (same amount of ACN as chloroform). The two solutions are combined and stirred for 1 h. Afterwards the electrolyte mixtures are dried in high vacuum at 80 °C for at least 15 h. After drying the electrolytes are transferred to a argon-filled glovebox for later use.

Ionic Conductivity: The ionic conductivity was evaluated by electrochemical impedance spectroscopy. For temperature-dependent ionic conductivity measurements, the temperature was controlled by a climate chamber (80 – 25 °C). For each temperature step, the impedance spectrum is recorded after an equilibration interval of 1.5 h. All electrochemical impedance spectroscopy (EIS) measurements were conducted in a frequency range of 3 MHz – 0.1 Hz using a sinusoidal excitation potential of 20 mV. EIS measurements were reconstructed by the extended distribution of relaxation times-(eDRT)-analysis using the software *ec–idea* provided by the Department of Electrical Energy Systems at the University of Bayreuth. Fitting parameters: Number of time constants: 159, 3-fold number of measured frequencies, extended upper time constant limit by 1 decade, extended lower time constant limit by 1 decade, regularization parameter: 0.1. The CR2032 coin cell setup consisted of lithium-blocking stainless steel electrodes and the respective electrolyte. The electrolyte geometry is ensured by a Telfon® spacer (500 μm thickness, diameter 16 mm, cavity diameter 3 mm). The spacer cavity is filled with electrolyte during cell preparation.

Linear Sweep Voltammetry: Linear sweep voltammograms were recorded at 70 °C and a scan rate of 0.1 mV s^{-1} between OCV (2.6 V vs. Li/Li^+) and 6 V. The CR2032 coin cell setup consisted of a lithium electrode (as reference and counter electrode) and a stainless-steel electrode (as working electrode). The electrolyte geometry is ensured by a Kapton® spacer (135 μm thickness, diameter 16 mm, cavity diameter 4 mm). The spacer cavity is filled with electrolyte during cell preparation.

3. Results and discussion

Synthesis of the linear and hyperbranched polymers are shown in Scheme 1. Tri-acrylate building block **M1** was reacted with 1-butanethiol to produce the di-acrylate monomers **M2** which was characterized by ^{1}H NMR (Figure S1). **M1** was chosen as the primary building block as it is readily available from commercial sources. Furthermore, the $–CH_2CH_3$ group attached to the central sp^3 hybridized carbon was anticipated to contribute to amorphous character of the resulting polymers with low glass transition temperature, which is desirable for the solid polymer electrolyte application. Polycondensation between **M2** and 1,3-propane dithiol or 1,6-hexane dithiol in presence of Me_2PPh catalyst produced linear polyesters **L-P1** and **L-P2**, respectively. Little amount of methyl acrylate was used as the chain stopper to avoid presence of free thiols at the chain-ends. For solid polymer electrolyte applications, it is desirable that the polymers exhibit low glass transition temperature (T_g). This is precisely the reason why a monomer like **M2** was chosen as it was anticipated that the pendant moieties would inhibit crystallization and help in achieving low T_g. **L-P1** and **L-P2** were synthesized to examine the effect of density of ester groups in the backbone on the ion conductivity as it would depend on the coordination of the ester carbonyl to the Li ion. To examine the effect of linear and hyperbranched polymers of similar nature, monomer **M1** was used to synthesize hyperbranched polymer by a A3 + B2 condensation approach [21–24] with the same two dithiols. For the synthesis of the hyperbranched polymers, tri-acrylate monomer **M1** was taken in little excess (**M1**: dithiol = 1.4: 1) to avoid the crosslinking-induced gelation during polymerization. This produced hyperbranched polymers **HB-P1a** and **HB-P2a** which were reacted with 1-butanethiol to consume the unreacted acrylate groups in the linear/ dendritic units of the hyperbranched polymers producing **HB-P1** and **HB-P2** (Scheme 1). Representative ^{1}H NMR spectra of **L-P2** and **HB-P2** are shown in Fig. 1. For the linear polymer **L-P2**, a singlet appears at $\delta = 4.03$ ppm, which can be assigned to Ha″ ($–CH_2$ proton) adjacent to the ester group and three clear triplets appear at $\delta = 2.4–2.8$ ppm which can be assigned to the methylene protons in the backbone (He, Hf, Hg, Fig. 1). Other methylene protons appear in the up-field region ($\delta = 1.4–1.6$ ppm, 0.8–1.0 ppm). For the hyperbranched polymer **HB-P2a**, multiple peaks appeared in the region of $\delta = 4.03$ ppm, which can be assigned to difference in the chemical environment of the $–OC\underline{H}_2$ protons (Ha′, Ha″) adjacent to the ester groups in the dendritic, terminal and linear units of the polymer. From the integration of the peaks corresponding to linear (L), terminal (T) and dendritic (D) units, the degree of branching (DB) of **HB-P1a** and **HB-P2a** was estimated to be 0.59 and 0.62, respectively, using the following equation.

$$DB = \frac{D + T}{D + T + L}$$

Peaks appearing in the region $\delta = 5.8–6.4$ ppm can be assigned to the unreacted acrylate groups (in the linear and terminal units). After the polymer was treated with 1-butanethiol, in the resulting polymer **HB-P2**, all the acrylate protons disappeared indicating complete consumption and interestingly multiple peaks appearing in the $\delta = 4.00$ ppm region due to the $–OC\underline{H}_2$ protons merged to almost a single peak similar to the linear polymer as in this case the difference in chemical environment of the different units in the hyperbranched polymer reduced significantly. Likewise, **L-P1** and **HB-P2** were also characterized by ^{1}H NMR spectra, which are shown in the supporting information (Figure S2-S4). Molecular weight of the polymers was estimated by size exclusion chromatography (Figure S5-S6), which revealed moderate values (M_n in the range of 5000–6000 g/mol, Đ < 1.3) for all four polymers (Table 1).

The synthesized polymers were evaluated for their potential as solid polymer electrolytes. The polymers were converted to solid polymer electrolytes by incorporating lithium bis(trifluoromethanesulfonyl) imide (LiTFSI) at a molar ratio of polymer carbonyl units to lithium of C=O/Li of 10. Given the variations in polymer architecture and chemical composition, this corresponded to salt concentrations ranging between 12 and 15 wt%, as detailed in Table S1. As the molar ratio of carbonyl units to lithium ions can significantly influence ionic conduction within the polymer matrix, [16,25] we opted for a consistent molar ratio rather than the same weight proportion. Thermal properties were evaluated by thermogravimetric analysis and differential scanning calorimetry. Values are summarized in Table 1. The polymers showed sufficient thermal stability for the application in solid-state lithium-metal batteries. The polymers showed a thermal decomposition onset of around 300 °C. **L-P1** showed an additional decomposition step starting at 160 °C, however terminal decomposition was comparable to the other three polymers (Fig. 2a).

Further, the thermal transitions of the polymers and corresponding electrolytes were investigated. All of the pristine polymers only showed a glass transition (Fig. 2b). The glass transition temperature (T_g) of the polymers with shorter spacer lengths (**L-P1** and **HB-P1**) showed a comparable T_g of around −45 °C. With longer spacer lengths in **L-P2** and **HB-P2** the glass transition shifted to lower temperatures of about −50 °C. Further spacing of the ester groups is commonly resulting in lower glass transition temperatures. The rational design and choice of spacer units between ester moieties was investigated by Rosenbach et al [16]. By changing the number of methylene units the glass transition temperature could be incrementally tuned from + 32 °C to −44 °C in methacrylate-based polymers with two ester groups as side chain modification [16].

Notably, both pristine polymers and corresponding electrolytes

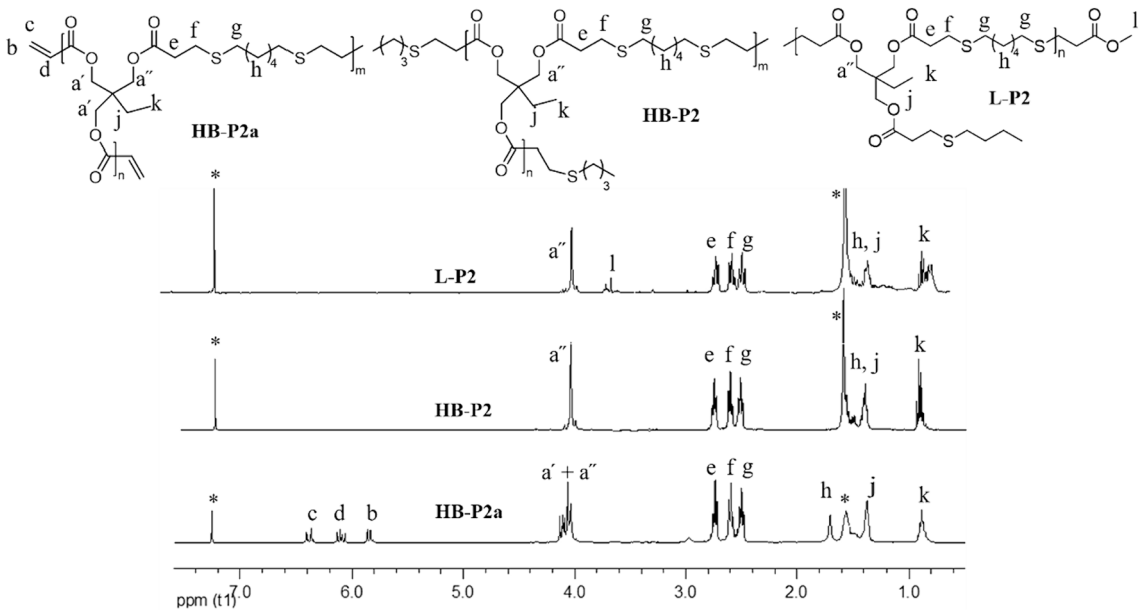


Fig. 1. Representative ^{1}H NMR spectra of a linear and hyperbranched polymer in $CDCl_3$. For the hyperbranched case, acrylate containing pre-polymer spectrum is also shown for comparison purposes. (* indicates residual solvent peaks).

Table 1
Overview of achieved molecular weights and polymer dispersity and thermal properties of the linear and hyperbranched polymers and corresponding electrolytes.

Polymer	M_n[a]	$Đ$[b]	$T_{d,onset}$	$T_{g,polymer}$[c,d]	$T_{g,electrolyte}$[c,e]
	g mol^{-1}		°C	°C	°C
L-P1	5500	1.20	329	−45	−36
L-P2	5000	1.21	348	−51	−43
HB-P1	5500	1.18	349	−46	−36
HB-P2	6300	1.25	354	−51	−40

[a] determined from SEC: THF eluent, PMMA standards.
[b] M_w/M_n.
[c] determined from the second heating cycle at 10 K min^{-1}.
[d] glass transition of the pristine polymer.
[e] glass transition of the polymer electrolyte (mixture of polymer + LiTFSI) at a C=O/Li ratio of 10.

exhibited no melting/crystallization peaks. The addition of LiTFSI resulted in an increase of the T_g by around 10 °C in all cases, which is commonly observed for solid polymer electrolytes.

This amorphous character is beneficial for the application as solid polymer electrolytes, as ion transport is usually observed in the amorphous polymer domains. The ionic conductivity (σ) was evaluated by electrochemical impedance spectroscopy (EIS) at different temperatures using ion-blocking stainless steel electrodes (StSt|electrolyte|StSt). The cell geometry enables the isolated evaluation of the ionic conductivity due to the absence of any charge-transfer processes. The recorded impedance data reveals one semicircle corresponding to the ionic conductivity of the polymer electrolytes and non-ideal capacitive behavior attributed to the electrode interfaces towards high Z' values and is well represented by the employed eDRT analysis (Fig. 3a, Figure S7-10). The evaluation of the Nyquist plots was performed by the extended distribution of relaxation times (eDRT) analysis. This enabled the evaluation of the EIS measurements without pre-defined equivalent circuits, as the analysis method uses no ab initio knowledge to reconstruct the

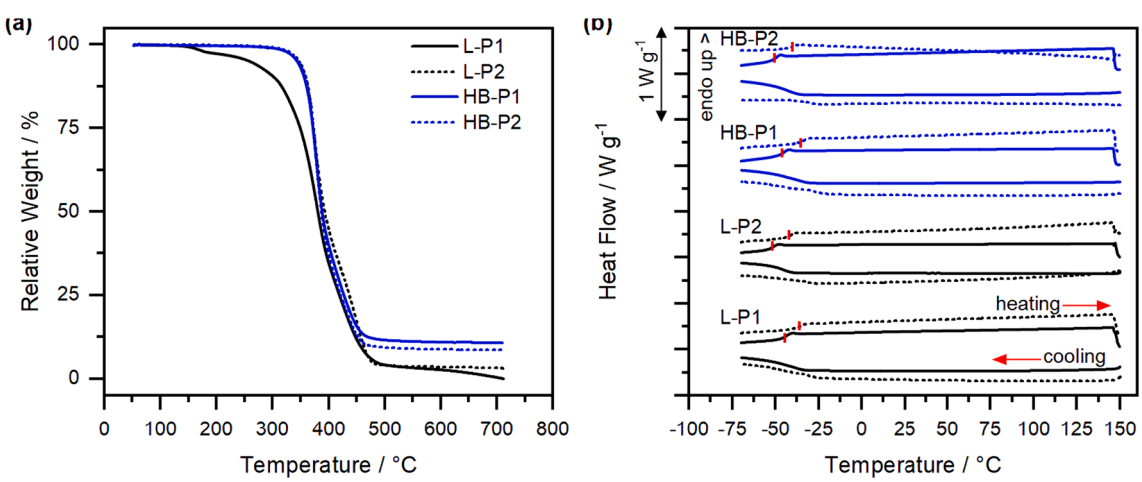


Fig. 2. Thermogravimetric analysis and differential scanning calorimetry of the (black lines) linear polymers **L-P1** and **L-P2**, and (blue lines) hyperbranched polymers **HB-P1** and **HB-P2**. (a) Thermal decomposition curves at a heating rate of 10 K min^{-1} under nitrogen atmosphere. (b) Thermal transitions of the (solid lines) pristine polymers and (dotted lines) corresponding electrolytes at a C = O/Li ratio of 10. The second cooling and heating cycle is shown for all samples. The glass transition temperature of the pristine polymers and electrolytes are marked with red indicators and were determined from the second heating curves.

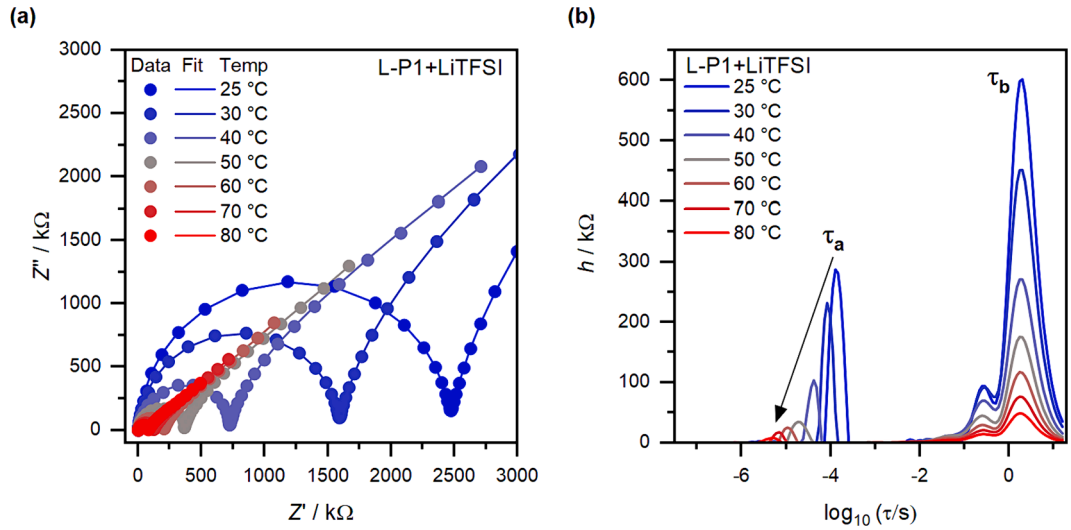


Fig. 3. (a) Exemplary Nyquist plot of temperature-dependent EIS measurements and eDRT fit for the L-P1 + LiTFSI (C=O/Li 10) electrolyte. (b) Extended distribution of relaxation times. The DRT results are represented by the absolute distribution function h(τ). τ_a and τ_b can be attributed to the ionic conductivity and capacitive behavior, respectively.

impedance data [26–28]. Exemplary data is shown for L-P1 in Fig. 3b and in Figure S11-13 for the remaining polymer electrolytes. The two processes (ionic conductivity and capacitive behavior) occur at different relaxation times. The relaxation time constant of the ionic conductivity contribution (τ_a) is highly dependent on the temperature and shifts to faster relaxation times as well as exhibits decreasing ohmic polarization. On the other hand, the corresponding relaxation time constant of the capacitive behavior (τ_b) is not shifting with temperature. The determined values of the ionic conductivity are summarized in Table 2.

The ionic conductivity of all of the four materials was found to be relatively similar with σ@25 °C between $1.5x10^{-7}$ and $2.6x10^{-7}$ S cm^{-1} and σ@70 °C between $4.54x10^{-6}$ and $3.59x10^{-6}$ S cm^{-1}. Notably, the electrolytes of linear polymers (**L-P1** and **L-P2**) have marginally higher ionic conductivity, especially at lower temperatures (Fig. 4a). Generally, the ionic conductivity can consist of intra- and interchain hopping events facilitated by low glass transition temperatures (or i.e. high polymer flexibility). Also, macroscopic polymer movements can contribute in a vehicular manner [29,30]. We hypothesize that the observed increase in ionic conductivity could originate from less restricted chain flexibility and higher polymer mobility provided in a linear polymer compared to hyper-branched polymer counterparts.

The ionic conductivity values of **L-P2** are compared to a polyester electrolyte consisting of linear poly(ε-caprolactone) (PCL) with a LiTFSI content of 19.5 wt% (C=O/Li = 10.4). The data for the whole temperature range of 25–80 °C is presented in Figure S14. Similar to PEO, PCL is a semicrystalline polymer with a melting point at about 50 °C. At high temperatures above the melting point of PCL, the ionic conductivity of PCL is higher than the values for the SPEs presented here. Below the melting point of PCL, its σ value decreases significantly by one order of magnitude and at 25 °C for a comparable salt concentration (C=O/Li = 10) similar ionic conductivities are obtained for **L-P2** and PCL electrolytes; the σ values being $2.6x10^{-7}$ and $3.2x10^{-7}$ S cm^{-1}, respectively. Thus, the crystallization-induced decrease of the ionic conductivity in linear PCL emphasizes the necessity of amorphous polymers as presented here. In further experiments, the salt content in linear and hyperbranched polymer electrolytes can be optimized to increase the ionic conductivity of these systems.

Table 2
Overview of the ionic conductivity and electrochemical stability of the polymers.

Polymer	σ @25 °C	σ @70 °C	$V_{d,LSV}$ [a]	$V_{d,LSV}$ [b]
	S cm^{-1}	S cm^{-1}	V	V
L-P1	$2.27x10^{-7}$	$4.54x10^{-6}$	3.78	4.39
L-P2	$2.58x10^{-7}$	$4.53x10^{-6}$	3.72	3.53
HB-P1	$1.79x10^{-7}$	$4.42x10^{-6}$	3.74	4.51
HB-P2	$1.54x10^{-7}$	$3.59x10^{-6}$	n.d.	n.d.

[a] Potential of the first decomposition onset.
[b] Potential of the second decomposition onset.

Lastly, we evaluated the oxidative stability of the polymer electrolytes against lithium-metal electrodes. The observed electrochemical stability windows were evaluated geometrically by fitting the oxidative current response (Fig. 4b). Thereby we distinguished between the first and second degradation onset. The determined onset potentials for the first degradation step were 3.78, 3.72, and 3.74 V for the polymers **L-P1**, **L-P2**, and **HB-P1**, respectively. LSV was not performed for **HB-P2**. The observed electrochemical stability windows are relatively low, as high-potential nickel-manganese-cobalt-oxide (NMC) cathodes operate at potentials above 4 V. However, the electrochemical stability may be adequate when paired with alternative cathode chemistries, such as in lithium-sulfur batteries (operating at around 2.3 V) or lithium–iron-phosphate-cathodes (3.4 V). Consequently, the application of polymer electrolytes in such batteries could be of future interest.

4. Conclusion

In this manuscript, we have shown the synthesis of new polyester-based materials for potential use as solid polymer electrolytes (SPE) in batteries. Polymers reported herein are particularly interesting considering the synthetic ease of these polyesters by 100 % atom-efficient thiol-acrylate Michael addition reaction, operating at room temperature. By choosing appropriate monomers, we could indeed make fully amorphous polyesters (linear or hyperbranched) with significantly low glass transition temperature, that is highly desirable for efficient ion conduction. However, in the present systems conductivity values were moderate, and oxidative stability of the polymer electrolytes against lithium-metal electrodes was just adequate. This underscores the need for further structural engineering with these new systems, which can be achieved easily due to the synthetic ease of this methodology. In further experiments, the salt content in linear and hyperbranched polymer electrolytes can be individually varied to obtain the optimum salt concentration towards best ionic conductivity of these systems. Furthermore, the different functional di-thiols could be of interest to introduce more ester or ether moieties for better ion coordination/conduction. Our

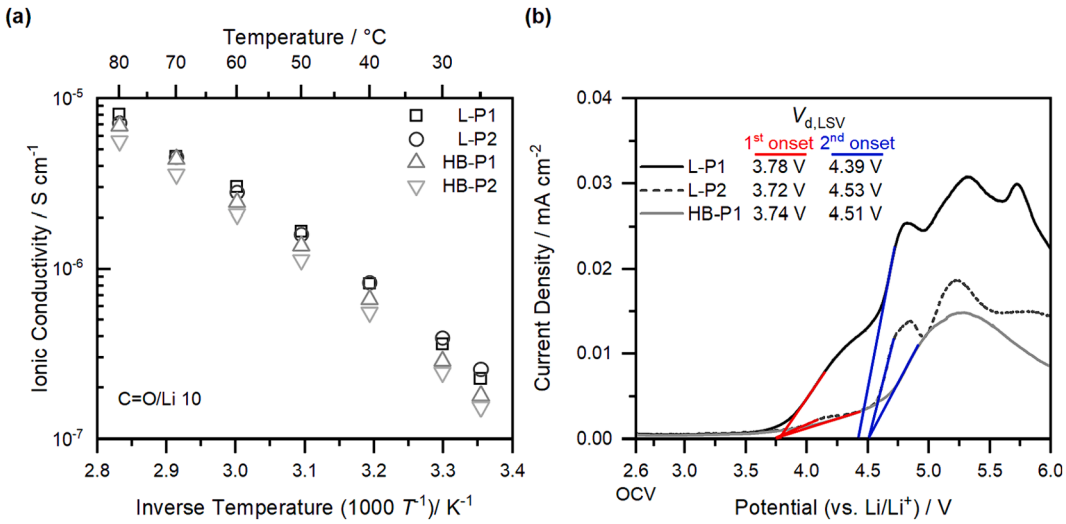


Fig. 4. (**a**) Arrhenius plots of the temperature-dependent ionic conductivity (80 °C – 25 °C). (**b**) Linear sweep voltammograms in Li|electrolyte|StSt cells at a sweep rate of 0.1 mV s^{-1} from OCV to 6 V. (red) First decomposition onset and (blue) second decomposition onset.

studies indicate linear polymers to be better candidates than hyperbranched analogues for this application. Synthesis of new polyester analogues with even lower T_g, higher conductivity, and oxidative stability and in-depth mechanistic study of ion conduction in these materials are underway in our laboratory.

CRediT authorship contribution statement

Jannik Petry: Writing – original draft, Visualization, Methodology, Investigation. **Krishna Dan:** Writing – original draft, Investigation. **Atish Nag:** Writing – original draft, Investigation. **Mukundan Thelakkat:** Writing – review & editing, Supervision, Conceptualization. **Suhrit Ghosh:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.eurpolymj.2024.112965.

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